

FIRST-STAGE ROCKET BODY STATE ESTIMATION WITH SPACE-BASED INFRARED SENSORS: FILTER COMPARISON AND SENSOR GEOMETRY ANALYSIS

Nathan W. O'Brien*

This paper presents an end-to-end simulation study of a disposable first-stage rocket body state estimation problem using two geostationary infrared (GEO IR) sensors in an angles-only measurement configuration. A two-stage dynamics model capturing behaviors during powered boost and ballistic coast is examined. A nine-state Singer-augmented model capturing position, velocity, and unknown non-gravitational acceleration is estimated with Extended Kalman (EKF), Unscented Kalman (UKF), and Interacting Multiple Model (IMM) filters. A quantification of how sensor placement affects depth observability reveals that, for the EKF, UKF, and IMM implementations studied here, triangulation geometry influences estimation accuracy more strongly than the choice of filter architecture. In the nominal 45° separation case, the EKF and UKF each achieve 1.621 km 3D position RMSE and 0.096 km/s 3D velocity RMSE. The IMM performs marginally better, achieving 1.607 km 3D position RMSE and 0.094 km/s 3D velocity RMSE, with all three filters yielding a mean normalized innovation squared near the expected value (3.59 vs. 3.64). The EKF and UKF produce numerically near-identical results, while the IMM delivers a smooth, probabilistically-weighted boost-to-coast phase transition that allows for marginal increases in filter accuracy. A line-of-sight (LOS) error decomposition shows that single-sensor depth uncertainty is the main source of large errors in the Earth-limb occlusion zone.

INTRODUCTION

Estimating the state of a rocket body's first-stage using space-based, angles-only measurements presents a fundamental observability challenge. During the boost phase, the vehicle undergoes sustained high acceleration under a thrust vector that is both large and rapidly varying, yet entirely unknown to any external observer. At burnout, it transitions abruptly to a ballistic coast arc governed by gravity alone. Infrared sensors aboard geostationary satellites observe the vehicle from a range of 35,786 km, though each sensor measures only the *angular* direction to the vehicle in azimuth and elevation. The result is a fundamentally underdetermined range problem. Angle measurements alone constrain two of three spatial degrees of freedom, leaving the depth (the distance along the sensor's line of sight) weakly observable through dynamical parallax alone.

Quantifying how sensor placement affects this depth observability is the central goal of this paper. Two geostationary sensors with adequate angular separation can triangulate the vehicle position in three dimensions, collapsing the depth ambiguity. However, the Earth's disk imposes a hard occlusion zone that can eliminate one sensor's field of view over portions of a trajectory, abruptly degrading tracking accuracy. Understanding where this occurs and why has direct implications for constellation design of early-warning systems.

*Graduate Student, School of Aerospace Engineering, Georgia Institute of Technology

Three filter architectures are evaluated: the Extended Kalman Filter (EKF), the Unscented Kalman Filter (UKF), and the Interacting Multiple Model (IMM) filter. All three use a Singer-augmented dynamics model that treats the unknown thrust as a colored-noise process, requiring no guidance-law information. A sensor longitude sweep across 72 geometries characterizes the accuracy and consistency of each filter as a function of sensor separation angle.

The principal contributions of this paper are: (1) a complete EKF/UKF/IMM implementation for angles-only two-phase rocket body state estimation with unknown non-gravitational acceleration; and (2) a LOS error decomposition that explains the range-blindness mechanism, the occlusion-onset overshoot, and the asymmetric filter recovery.

SCENARIO AND TRUTH MODEL

Trajectory Geometry

The simulated scenario represents a first-stage rocket body launched from China’s Jiuquan Satellite Launch Center (40.96°N , 100.29°E) on a southeast-heading trajectory. The vehicle’s engine burns for 320 s, reaching a peak altitude of approximately 100 km at 1,200 km downrange, then coasts ballistically to an ocean impact point at approximately 12°N , 141°E , east of the Philippine archipelago in the western Pacific. Throughout flight, the first-stage travels 5399.758 km downrange in 876.728 s.

The truth trajectory is propagated in a fixed inertial plane determined by the launch site and flight azimuth. This plane is defined by the launch-site radius vector and the initial velocity direction, and all thrust is contained within it. Figure 1 shows the trajectory in Earth-Centered, Earth-Fixed (ECEF) coordinates alongside the nominal GEO sensor constellation.

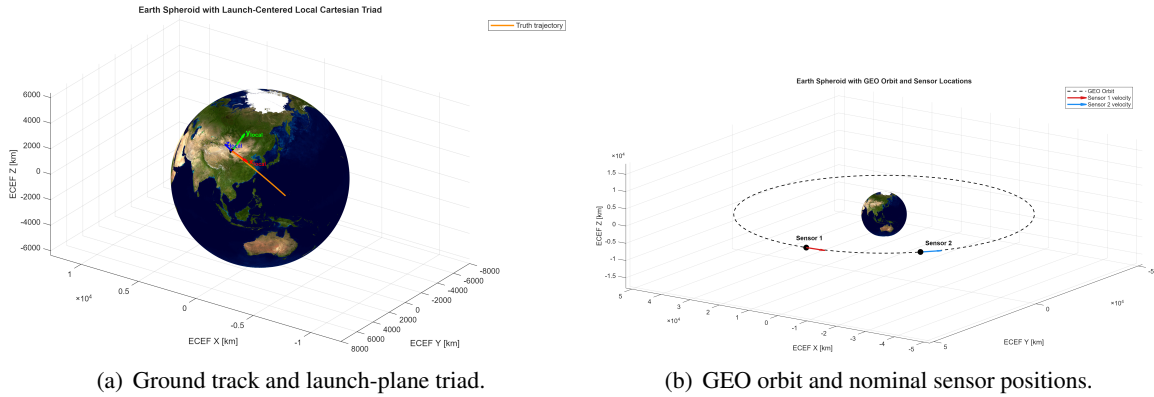


Figure 1. Scenario geometry. Left: Truth Trajectory. Right: GEO-1 (red) is fixed at 93.8°E ; GEO-2 (black) is at 139.0°E for the nominal case, giving a 45.2° sensor separation.

Nominal Scenario Rationale

The nominal sensor positions were chosen to match two satellites in the SBIRS GEO network, which fields assets in this longitude band to cover the eastern Pacific launch corridor.¹ A secondary motivation was to examine whether the geometry sweep can offer a purely estimation-theoretic commentary on why a particular angular separation between two such satellites performs well, acknowledging that constellation design is driven by many factors beyond tracking accuracy alone.

The vehicle parameters are chosen to be semi-realistic for a Chinese orbital-class first stage. A sustained thrust-to-weight ratio of $2.525g$, burnout at $t = 320$ s, and a peak altitude of approximately 100 km are broadly consistent with publicly available performance envelopes for Long March-class launch vehicles.² They also remain simplified enough to support a closed-form Pontryagin-optimal steering solution for truth generation. Together, these choices exercise both the boost-phase filter architectures and the full range of occlusion geometries explored in the geometry sweep.

Boost Steering Law

The boost trajectory is driven by an optimal thrust steering program derived from Pontryagin’s Minimum Principle (PMP), solved offline as a boundary value problem. The commanded pitch angle $\beta_{\text{cmd}}(t)$ decreases in a mildly non-linear manner from $+40^\circ$ at launch to -9° at burnout, tilting the thrust vector over time to maximize downrange energy injection.

This steering law is used *only* for truth generation in the nominal case. The filters have no access to it. The thrust direction and magnitude are treated as entirely unknown and must be inferred from measurements alone.

Truth Propagation and Disturbance Model

The truth state is propagated numerically on a $\Delta t_{\text{truth}} = 0.1$ s time grid within the fixed launch plane. The total non-gravitational acceleration is:

$$\mathbf{a}_{\text{ng}} = \mathbf{a}_{\text{thrust}}(t) + \boldsymbol{\eta}(t), \quad (1)$$

where $\boldsymbol{\eta}$ is a Gauss-Markov disturbance injected in the in-plane tangential and radial directions; the truth trajectory has no independent cross-range forcing and remains confined to the launch plane. Boost-phase disturbance parameters are $\tau_{\text{boost}} = [4, 6]$ s and $\sigma_{\text{boost}} = [0.40, 0.10]$ m/s² for tangential and radial components, respectively. Coast-phase parameters are $\tau_{\text{coast}} = 10$ s and $\sigma_{\text{coast}} = 0.02$ m/s² in both in-plane directions.

Sensor Model

Two zero-inclination GEO IR sensors are modeled at fixed equatorial longitudes. Each sensor reports azimuth and elevation at 1 Hz in the local East-North-Up (ENU) frame, with independent Gaussian noise $\sigma_{\text{az}} = \sigma_{\text{el}} = 0.01^\circ$ per axis. A 10% independent dropout probability per sensor per epoch simulates link-layer outages. Detection is triggered when the vehicle first exceeds 7 km altitude, used here as a simple proxy for the onset of reliable space-based infrared visibility.¹ An Earth-limb blockage check prohibits measurements for which the GEO-to-vehicle LOS intersects the solid Earth ($R_{\oplus} = 6378$ km), which is the source of the occlusion zones analyzed in Section .

Figure 2 shows the number of sensors reporting at each epoch for the nominal 45° geometry. Dual-sensor epochs account for approximately 79% of the total trajectory. The remainder is split between single-sensor and rare simultaneous-dropout epochs.

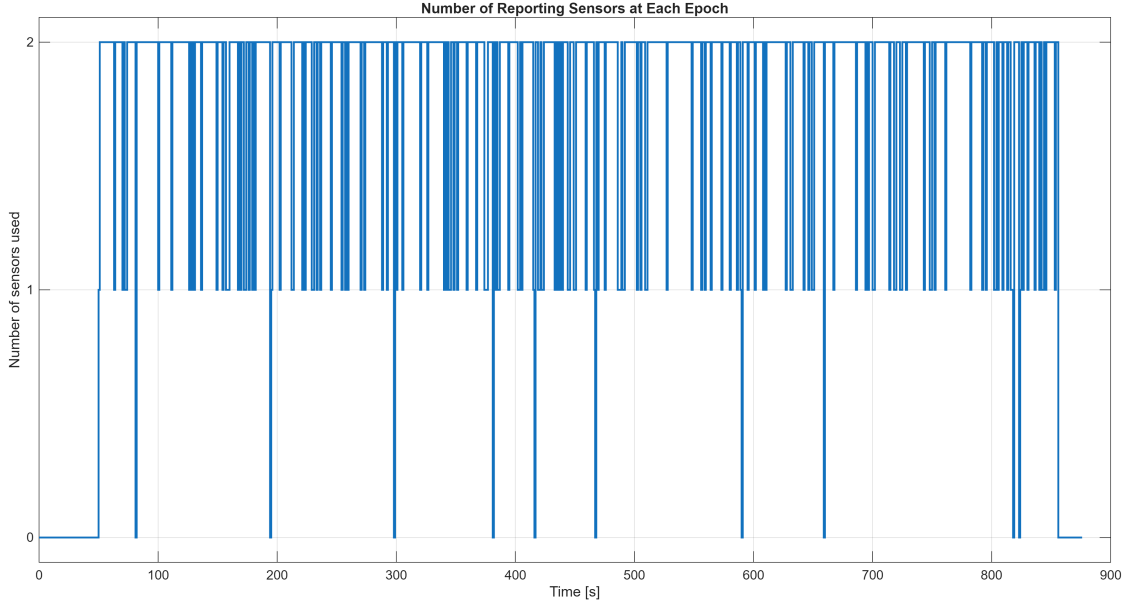


Figure 2. Number of reporting sensors per epoch at the nominal 45° geometry. Dropout is independent across sensors; the no-measurement epochs arise from simultaneous dropout, occurring at the expected 1% rate. Note: there are no measurements at the beginning or end of the trajectory because the vehicle is below 7 km altitude.

FILTER ARCHITECTURES

State Vector and Dynamics

All three filters share the same nine-dimensional state vector:

$$\mathbf{x} = \begin{bmatrix} \mathbf{r} \\ \mathbf{v} \\ \mathbf{a} \end{bmatrix} \in \mathbb{R}^9, \quad (2)$$

where $\mathbf{r} \in \mathbb{R}^3$ is position in Earth-Centered Inertial (ECI) coordinates, $\mathbf{v} \in \mathbb{R}^3$ is ECI velocity, and $\mathbf{a} \in \mathbb{R}^3$ is the unknown non-gravitational acceleration (thrust plus disturbances). The continuous-time dynamics are:

$$\dot{\mathbf{r}} = \mathbf{v}, \quad \dot{\mathbf{v}} = \mathbf{g}(\mathbf{r}) + \mathbf{a}, \quad \dot{\mathbf{a}} = -\frac{1}{\tau}\mathbf{a} + \boldsymbol{\eta}, \quad (3)$$

where $\mathbf{g}(\mathbf{r}) = -(\mu_\oplus/|\mathbf{r}|^3)\mathbf{r}$ is point-mass gravitational acceleration with $\mu_\oplus = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$.

The Singer Acceleration Model

The Singer model was chosen because thrust is physically real and temporally correlated, yet entirely unknown to the observer. A white-noise model would be too optimistic about how rapidly thrust can change, and an explicit thrust model is impossible without access to the guidance law. The acceleration state $\mathbf{a}$ is therefore modeled as a first-order Gauss-Markov process.³ The autocorrelation of each component decays exponentially: $C_a(\delta) = \sigma^2 e^{-|\delta|/\tau}$, where τ is the correlation time and σ is the steady-state RMS acceleration. The stochastic driving term satisfies $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, (2\sigma^2/\tau)\mathbf{I})$.

This model captures two physically important properties: thrust is not impulsive (the correlation time τ sets the timescale over which the acceleration direction changes), and it is not constant. The filter expects $\mathbf{a}$ to slowly wander, allowing it to track the continuously-pitching PMP steering law without being fed the steering commands directly.

Two parameter sets cover the two flight phases:

- *Boost*: $\tau_b = 180$ s, $\sigma_b = 20$ m/s². The long correlation time reflects the slowly-varying PMP pitch program, and the large sigma covers a $\sim 2g$ sustained thrust in any direction.
- *Coast*: $\tau_c = 10$ s, $\sigma_c = 0.05$ m/s². The short correlation time and small sigma model rapidly decorrelating residual disturbances (atmospheric drag at high altitude, J_2 perturbations, etc.).

The model is applied *anisotropically* to exploit the known launch-plane geometry. The in-plane Singer parameters take the values above. The out-of-plane sigma is reduced to $\sigma_{\perp,b} = 0.5$ m/s² during boost and $\sigma_{\perp,c} = 0.01$ m/s² during coast. The PMP steering law constrains all thrust to the fixed inertial launch plane, so out-of-plane acceleration is expected to be small. The non-zero out-of-plane sigma prevents the cross-plane block of the covariance from becoming singular.

Discrete-Time Process Noise: The Van Loan Method

The continuous-time process noise density matrix is:

$$\mathbf{Q}_c = \begin{bmatrix} \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \mathbf{0}_3 \\ \mathbf{0}_3 & \mathbf{0}_3 & \Sigma_a \end{bmatrix}, \quad (4)$$

where $\Sigma_a = \mathbf{U} \text{diag}(q_{\parallel}, q_{\parallel}, q_{\perp}) \mathbf{U}^T$ is the in-plane/out-of-plane noise covariance rotated to ECI, with $q_{\parallel} = 2\sigma^2/\tau$ and $q_{\perp} = 2\sigma_{\perp}^2/\tau$.

Computing the exact discrete-time process noise covariance $\mathbf{Q}_d$ from $\mathbf{Q}_c$ requires evaluating an integral that does not have a closed form in general. The Van Loan method³ avoids this by forming a $2n \times 2n$ auxiliary matrix and computing a single matrix exponential:

$$\exp\left(\begin{bmatrix} -\mathbf{A} & \mathbf{Q}_c \\ \mathbf{0} & \mathbf{A}^T \end{bmatrix} \Delta t\right) = \begin{bmatrix} \mathbf{M}_{11} & \mathbf{M}_{12} \\ \mathbf{0} & \mathbf{M}_{22} \end{bmatrix}, \quad (5)$$

where $\mathbf{A}$ is the linearized continuous-time dynamics matrix from Equation (3). The discrete-time state transition matrix and process noise covariance are then extracted as:

$$\mathbf{F} = \mathbf{M}_{22}^T, \quad \mathbf{Q}_d = \mathbf{M}_{22}^T \mathbf{M}_{12}. \quad (6)$$

This approach is exact to machine precision and avoids the truncation errors that arise from naive Taylor-series approximations of $\mathbf{Q}_d$ at short time steps.

Extended Kalman Filter

The EKF linearizes the nonlinear dynamics and measurement functions about the current estimate.³ The prediction step propagates the state mean by numerical integration of Equation (3) over the 1 s measurement interval and propagates the covariance via:

$$\mathbf{P}_k^- = \mathbf{F}_k \mathbf{P}_{k-1}^+ \mathbf{F}_k^T + \mathbf{Q}_{d,k}, \quad (7)$$

where $\mathbf{F}_k = \partial f / \partial \mathbf{x} |_{\hat{\mathbf{x}}_{k-1}^+}$ is the state transition matrix evaluated at the previous posterior estimate.

The measurement update uses the Joseph stabilized form of the covariance equation to preserve positive-definiteness under finite-precision arithmetic:

$$\mathbf{K}_k = \mathbf{P}_k^- \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_k^- \mathbf{H}_k^T + \mathbf{R}_k)^{-1}, \quad (8)$$

$$\mathbf{P}_k^+ = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_k^- (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)^T + \mathbf{K}_k \mathbf{R}_k \mathbf{K}_k^T. \quad (9)$$

Flight-phase detection is fully autonomous. No burnout time is provided to the filter. The EKF monitors the norm of the estimated acceleration state $\|\hat{\mathbf{a}}\|$ and latches from boost to coast Singer parameters when $\|\hat{\mathbf{a}}\|$ falls below a threshold of $g_0/2 \approx 4.9 \text{ m/s}^2$, subject to the guard condition that thrust has already been detected ($\|\hat{\mathbf{a}}\| > g_0/2$ at some prior epoch). The latch is one-way. Once the filter commits to coast parameters, boost parameters are never reinstated. The thrust level and direction are never provided to the filter.

Unscented Kalman Filter

The UKF replaces Jacobian linearization with a deterministic sigma-point approximation.³ For a state of dimension $n = 9$ with scaling parameter $\lambda = \alpha^2(n + \kappa) - n$, a set of $2n + 1$ sigma points is drawn symmetrically from the prior covariance:

$$\chi_0 = \hat{\mathbf{x}}_k^-, \quad \chi_i = \hat{\mathbf{x}}_k^- + \left(\sqrt{(n + \lambda) \mathbf{P}_k^-} \right)_i, \quad \chi_{n+i} = \hat{\mathbf{x}}_k^- - \left(\sqrt{(n + \lambda) \mathbf{P}_k^-} \right)_i, \quad (10)$$

for $i = 1, \dots, n$. Each sigma point is propagated through the nonlinear dynamics and measurement functions, and the predicted mean and covariance are recovered as weighted sums.

Because the azimuth/elevation measurement functions are smooth and their curvature is small relative to the filter's operating uncertainty, the UKF and EKF produce numerically near-identical results on this problem. The UKF is implemented as an independent verification. All Singer parameters, initial conditions, and gating thresholds are shared identically between the two filters.

Interacting Multiple Model Filter

The IMM maintains $M = 2$ parallel Singer-EKF sub-filters, one tuned for boost and one for coast, and combines their estimates probabilistically.³ This replaces the EKF's one-way acceleration-norm latch with a smooth, measurement-driven phase transition. Each filter cycle proceeds in four steps.

Step 1: Interaction. The predicted probability that model j is active at step k is:

$$\bar{c}_j = \sum_{i=1}^M p_{ij} \mu_i^{k-1}, \quad (11)$$

where p_{ij} is the Markov transition probability from model i to model j . The mixing weights $\mu_{i|j}^{k-1} = p_{ij} \mu_i^{k-1} / \bar{c}_j$ are used to blend the sub-filter state estimates into a single mixed initial condition for each sub-filter:

$$\hat{\mathbf{x}}_{j,0} = \sum_i \mu_{i|j}^{k-1} \hat{\mathbf{x}}_{i,k-1}^+. \quad (12)$$

Step 2: Mode-Conditioned Filtering. Each sub-filter performs a standard EKF predict-and-update cycle from its mixed initial condition, using its own Singer parameters.

Step 3: Model Probability Update. The Gaussian likelihood of the current measurement under each model is:

$$\Lambda_j = \mathcal{N}(\mathbf{z}_k; \hat{\mathbf{z}}_j, \mathbf{S}_j), \quad (13)$$

and the model probabilities are updated and normalized:

$$\mu_j^k = \frac{\Lambda_j \bar{c}_j}{\sum_{\ell} \Lambda_{\ell} \bar{c}_{\ell}}. \quad (14)$$

Step 4: Estimate Fusion. The final estimate is the probability-weighted sum of sub-filter outputs:

$$\hat{\mathbf{x}}_k = \sum_j \mu_j^k \hat{\mathbf{x}}_{j,k}^+, \quad \mathbf{P}_k^+ = \sum_j \mu_j^k \left[\mathbf{P}_{j,k}^+ + \left(\hat{\mathbf{x}}_{j,k}^+ - \hat{\mathbf{x}}_k \right) \left(\hat{\mathbf{x}}_{j,k}^+ - \hat{\mathbf{x}}_k \right)^T \right]. \quad (15)$$

Markov Transition Matrix. The 2×2 transition matrix governs the flow of probability mass between models:

$$\mathbf{\Pi} = \begin{bmatrix} 0.9966 & 0.0034 \\ 0.0000 & 1.0000 \end{bmatrix}, \quad (16)$$

where rows index the current model (boost, coast) and columns index the next. Two deliberate design choices are encoded.

First, the coast model is made *absorbing* ($p_{21} = 0$): a disposable solid- or liquid-fueled first stage does not (typically) re-ignite after burnout, so the probability assigned to the coast model can only grow, never flow back to boost. This ensures that once the filter commits to coast, the large boost-phase process noise covariance $\mathbf{Q}_{d,\text{boost}}$ no longer contaminates the fused covariance, keeping the state estimate tight through the coast arc.

Second, the boost self-transition probability $p_{11} = 0.9966$ is calibrated so that the expected boost dwell time, $1/(-\ln p_{11}) \approx 294$ s, is comparable to the nominal burnout time of 320 s. This gives the filter broad coverage of possible burnout epochs without committing prematurely. A numerical floor of 10^{-6} is enforced on all model probabilities to keep every sub-filter numerically alive.

OBSERVATION MODEL

The measurement function $h_i(\mathbf{x})$ for sensor i maps inertial position into the sensor's local East-North-Up (ENU) frame in three steps: ECI-to-ECEF rotation, subtraction of the sensor's fixed ECEF position to form the LOS vector, and projection of that LOS vector into the sensor's local frame:

$$\mathbf{r}_{\text{ecef}} = \mathbf{R}_z(\omega_e t) \mathbf{r}, \quad \boldsymbol{\rho} = \mathbf{r}_{\text{ecef}} - \mathbf{r}_{s,i}, \quad \boldsymbol{\rho}_{\text{enu}} = \mathbf{C}_{\text{enu},i} \boldsymbol{\rho}, \quad (17)$$

where $\mathbf{R}_z(\omega_e t)$ is the ECI-to-ECEF rotation matrix with $\omega_e = 7.292 \times 10^{-5}$ rad/s, $\mathbf{r}_{s,i}$ is the sensor's fixed ECEF position, and $\mathbf{C}_{\text{enu},i}$ is the sensor-specific ECEF-to-ENU direction-cosine matrix. The measurements follow as:

$$\text{az}_i = \text{atan2}(\rho_e, \rho_n), \quad \text{el}_i = \text{atan2}\left(\rho_u, \sqrt{\rho_e^2 + \rho_n^2}\right). \quad (18)$$

The measurement noise for each sensor is $\mathbf{R}_i = \text{diag}(\sigma_{\text{az}}^2, \sigma_{\text{el}}^2)$ with $\sigma_{\text{az}} = \sigma_{\text{el}} = 0.01^\circ = 1.745 \times 10^{-4}$ rad. The measurement Jacobian $\mathbf{H}_{i,k} = \partial h_i / \partial \mathbf{r}|_{\hat{\mathbf{x}}_k^-}$ is computed analytically for the EKF

by differentiating Equation (18) through the coordinate chain and is nonzero only in the position columns of the full $\partial h_i / \partial \mathbf{x}$ matrix. The IMM sub-filters use the same measurement function but evaluate the Jacobian numerically with a central-difference approximation.

Measurement gating. Each sensor’s two-dimensional innovation $\boldsymbol{\nu}_{i,k}$ is validated independently with a χ^2 gate. The normalized innovation squared $\epsilon_{i,k} = \boldsymbol{\nu}_{i,k}^T \mathbf{S}_{i,k}^{-1} \boldsymbol{\nu}_{i,k}$ is compared to $\chi_{0.999}^2(2) = 13.82$, giving a 0.1% per-sensor false alarm rate. A sensor that fails the gate is excluded from the update for that epoch.

NOMINAL PERFORMANCE

The nominal scenario places GEO-2 at 139°E (45.2° separation from GEO-1). The filter is initialized at the known launch site with position uncertainty $\sigma_{r_0} = 20$ m, velocity uncertainty $\sigma_{v_0} = 50$ m/s, and acceleration uncertainty $\sigma_{a_0} = 20$ m/s². In this section, RMSE is defined over the tracked window from the first measurement epoch to the last measurement epoch.

Filter Accuracy

Figure 3 shows the estimated trajectory for all three filters overlaid on the truth in the launch plane. All three estimates track the truth closely at this scale, confirming rapid post-detection convergence and accurate tracking through both the boost arc and the ballistic coast.

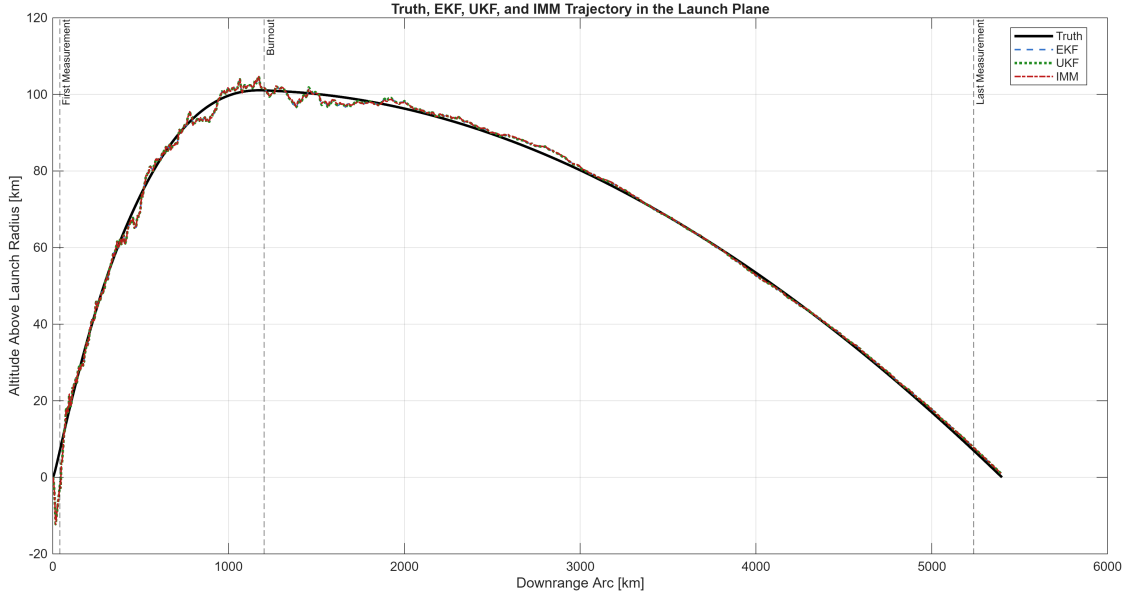


Figure 3. Truth (black solid), EKF (blue dashed), UKF (green dotted), and IMM (red dash-dot) trajectories in the launch plane. Vertical lines mark first measurement, burnout, and last measurement epochs.

Figure 4 shows the ECI position error for all three filters overlaid. The behavior follows three phases. Before detection ($t < 50$ s), the filter propagates in prediction-only mode. The 3σ covariance bounds grow as process noise accumulates. At detection, the first angle measurements arrive and the filter must reconcile a 20 m position prior with an angular observation from 35,786 km that constrains only two degrees of freedom. The 3σ bounds spike to ± 75 –100 km before the filter

converges. This convergence is rapid, within approximately 10–15 s of the first epoch, as errors collapse from the initial transient and remain bounded within the 3σ envelope, reaching an 3D position RMSEs of 1.621 km (EKF/UKF) and 1.607 km (IMM), as summarized in Table 1.

Post-convergence, all three filters behave nearly identically. The EKF and UKF differ only negligibly on all three axes, confirming that the measurement nonlinearity is insufficient to differentiate the two methods for this problem. The IMM shows only a marginal improvement in position RMSE.

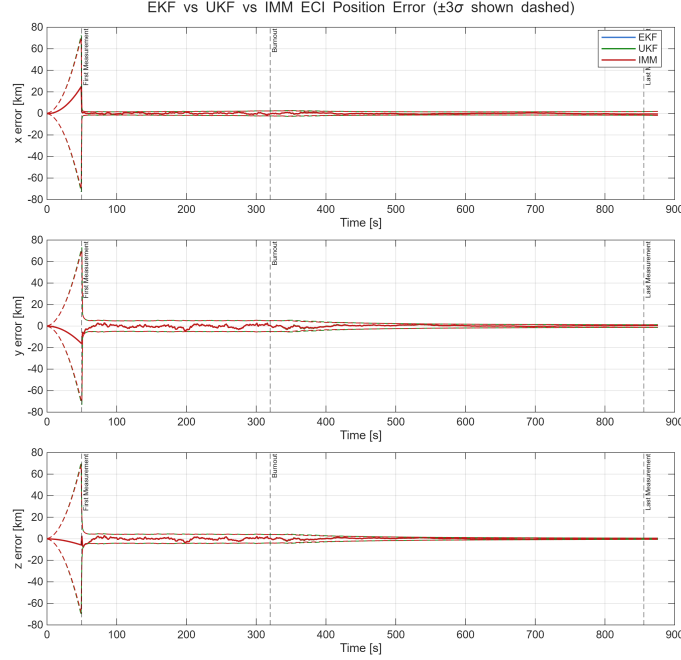


Figure 4. ECI position error (per axis) for EKF (blue), UKF (green), and IMM (red) at the nominal 45° geometry. Dashed lines show $\pm 3\sigma$ bounds for each filter. EKF and UKF differ only negligibly; IMM offers a marginal position RMSE improvement. All three filters are visually indistinguishable in both position and velocity error.

Figure 5 shows the corresponding per-axis ECI velocity error. All three filters are indistinguishable on every axis, with the IMM overlaid directly on the EKF and UKF throughout both boost and coast. The slight theoretical increase in velocity noise introduced by IMM model mixing is not discernible at this scale.

Table 1 summarizes overall estimation performance for all three filters across the tracked window from the first measurement epoch to the last measurement epoch. The EKF and UKF are numerically near-identical on every metric except complexity. The IMM offers a marginal position RMSE improvement of 0.9% and a marginal velocity RMSE improvement of 2.1%.

Table 1. Overall Filter Performance Summary: Nominal 45° Separation

Filter	3D Pos RMSE [km]	Mean Pos 1σ [km]	3D Vel RMSE [km/s]	Mean Vel 1σ [km/s]	Mean NIS (exp. 3.636)
EKF	1.621	1.544	0.096	0.088	3.591
UKF	1.621	1.544	0.096	0.088	3.591
IMM	1.607	1.523	0.094	0.085	3.590

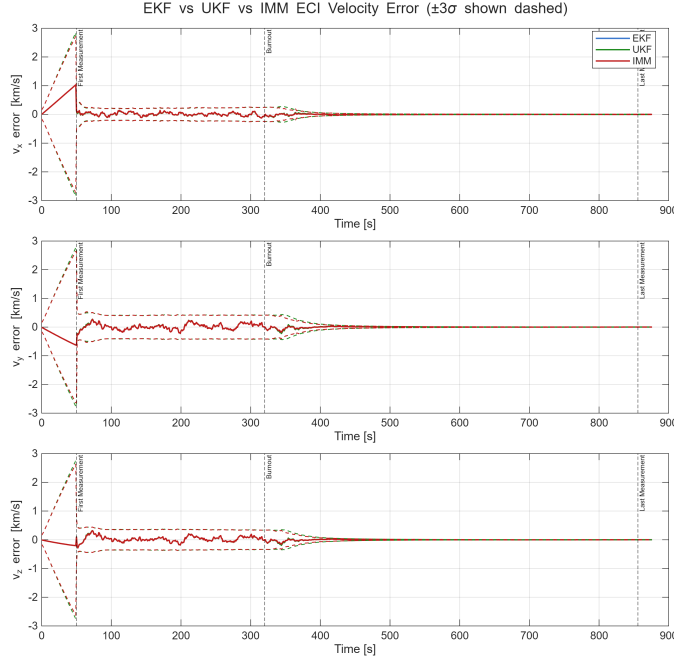


Figure 5. ECI velocity error (per axis) for EKF (blue), UKF (green), and IMM (red) at the nominal 45° geometry. Dashed lines show $\pm 3\sigma$ bounds. EKF and UKF differ only negligibly on all axes throughout boost and coast.

Filter Consistency

Figure 6 (left) shows the EKF normalized innovation squared (NIS). A well-tuned filter produces $\epsilon_k \sim \chi^2(m_k)$ where m_k is the number of measurements at epoch k . The mean NIS should equal the expected number of measurements per epoch. Here, mean NIS is reported over epochs with at least one accepted measurement, and the expected value is the corresponding mean measurement dimension. A mean NIS of 3.59 against the expected value of 3.636, a 1.3% shortfall, confirms that the filter covariance is well-calibrated throughout the trajectory. The handful of epochs exceeding the 95% chi-squared bound (red dashed line, $m=4$: $\chi_{0.975}^2(4) = 11.14$) are consistent with the expected 5% tail rate. The UKF and IMM NIS are numerically identical to the EKF.

IMM Phase Transition

Figure 6 (right) shows the IMM model probability evolution. The boost model probability μ_{boost} oscillates between 0.80 and 0.97 during boost, driven by measurement-dropout-induced variability in the model likelihood ratio. At true burnout ($t = 320$ s), $\mu_{\text{boost}} = 0.941$. The coast model probability then rises rapidly as the inferred acceleration $\hat{a}$ decays toward zero under the coast Singer model. The two probabilities cross at $t = 345.3$ s (25.3 s post-burnout), and μ_{coast} saturates to within 1% of unity at $t = 389.5$ s.

This 80–85 s transition lag is set by the Markov transition rate $p_{12} = 0.0034/\text{s}$: the probability mass can only shift as fast as the Markov chain and the likelihood ratio allow. The EKF’s acceleration-norm latch, by comparison, commits within 5–10 s of true burnout once $\|\hat{a}\|$ drops below threshold, at the cost of a one-step covariance discontinuity when the Singer parameters change. The IMM’s soft blend across sub-filters avoids this discontinuity throughout the transition.

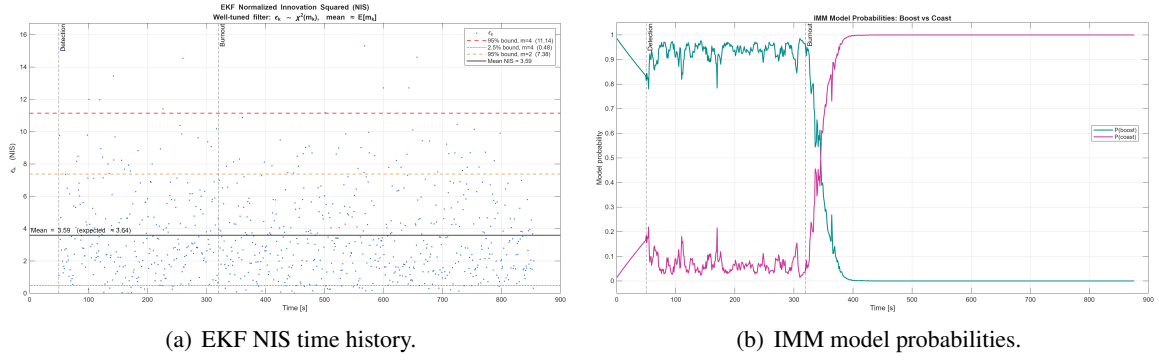


Figure 6. Filter consistency and IMM phase detection at the nominal geometry. Left: EKF NIS scatter with χ^2 bounds; mean 3.59 vs. expected 3.636. Right: IMM model probability evolution; μ_{coast} saturates to within 1% of unity 70 s after true burnout (dashed line).

SENSOR GEOMETRY AND OBSERVABILITY

Sweep Configuration

GEO-1 is held fixed at 93.8°E. GEO-2 is stepped through 72 separation angles from 0° to 355° in 5° increments. At each geometry, the EKF and IMM are run from launch to first-stage impact and the UKF is omitted because it is numerically identical to the EKF. Reported RMSE metrics are averaged over the tracked window from the first measurement epoch to the last measurement epoch.

Position Accuracy versus Separation Angle

Figure 7 shows 3D position RMSE versus separation angle. Five regimes are evident.

Near-coincident (0°–5°): GEO-2 is essentially co-located with GEO-1. The two sensors provide nearly redundant LOS vectors, and RMSE roughly approaches a value slightly below the single-sensor equilibrium value as the separation angle approaches 0°.

Dual-sensor (10°–85°): Both sensors provide geometrically independent LOS vectors. RMSE drops below 2.0 km and is largely insensitive to the exact separation within this band. The LOS intersection angle (Figure 9, left) grows proportionally with separation, providing progressively stronger depth triangulation. The nominal 45° geometry lies within this well-performing regime.

Occlusion onset (90°–120°): GEO-2 enters partial Earth-limb occlusion, losing visibility over progressively larger fractions of the trajectory as separation increases. RMSE rises sharply as more of the trajectory is occluded over this transition band.

Full occlusion (120°–275°): GEO-2 is blocked for the majority of tracked epochs (Figure 8, right). The filter operates almost entirely in single-sensor mode and RMSE saturates at a flat plateau.

Recovery (275°–355°): GEO-2 re-enters the FOV and RMSE collapses back to near-zero far more abruptly than the rise during occlusion onset. This asymmetry is explained in Section .

Filter Consistency Across Geometries

Figure 8 (left) shows the overall NIS ratio (mean NIS divided by expected NIS) at each separation angle. The ratio remains within about $\pm 4.5\%$ of unity across all 72 geometries, typically

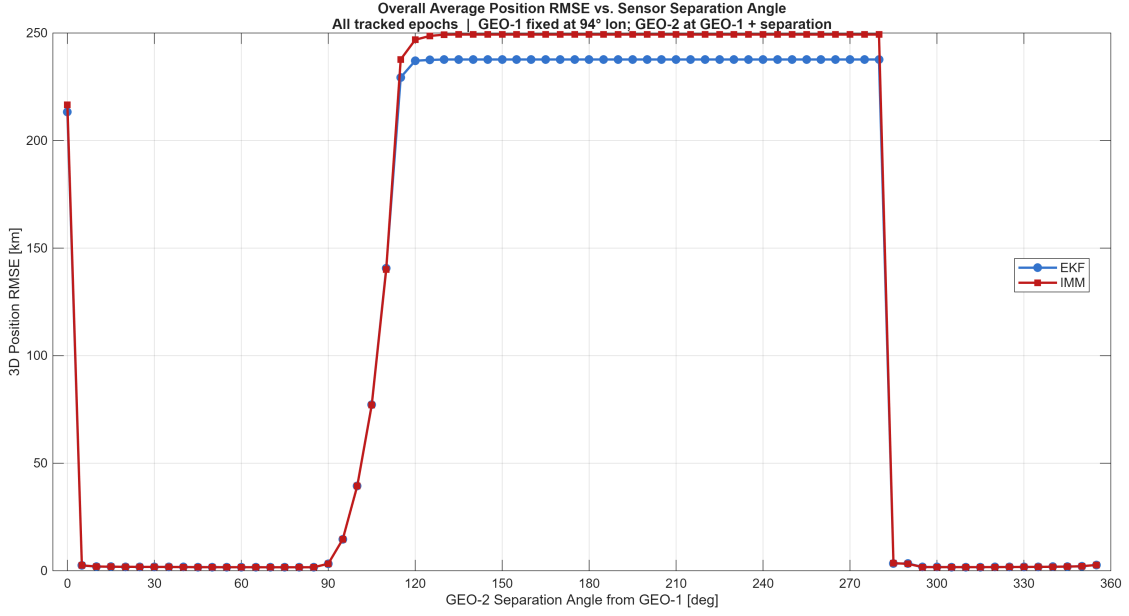


Figure 7. 3D position RMSE versus GEO-2 separation angle. The occlusion zone (120°–275°) drives RMSE to a 237–249 km plateau. EKF and IMM plateaus differ by 12 km owing to model-mixing noise in the depth direction (Section). Nominal 45° geometry falls in the optimal band.

sitting just below 0.98 and dipping to about 0.955 in the worst occlusion cases. This indicates mild underconfidence overall, with the IMM usually somewhat closer to unity except in the occlusion zone. A narrow spike to 1.028 appears at 110°, coinciding with the occlusion transition where GEO-2’s availability changes most rapidly and the filter’s expected measurement count is least stable. This result demonstrates that the Singer process-noise tuning maintains covariance calibration across dramatically different measurement geometries.

LINE-OF-SIGHT ERROR DECOMPOSITION

Range Blindness

The large single-sensor RMSE in the occlusion zone has a geometric root cause. With two sensors, the intersection of two independent LOS vectors triangulates the vehicle position in three dimensions and strongly constrains range. With only one sensor, azimuth and elevation constrain two angular degrees of freedom, but the distance along the LOS (the depth) must be recovered through dynamical parallax: the slowly-accumulating angular shift driven by the sensor’s own orbital motion (3.1 km/s at GEO altitude) and the vehicle’s changing trajectory. For a 320 s boost arc observed from 35,786 km, this parallax signal is weak.

To isolate how much of the total position RMSE comes from depth uncertainty, the position error vector $e_k = \hat{r}_k - r_{\text{truth},k}$ is decomposed into components parallel and perpendicular to the active sensor’s LOS unit vector $\hat{l}_k$:

$$e_{\parallel,k} = (e_k \cdot \hat{l}_k), \quad e_{\perp,k} = e_k - e_{\parallel,k} \hat{l}_k. \quad (19)$$

The cross-LOS component $|e_{\perp}|$ is directly constrained by the angle measurements and remains

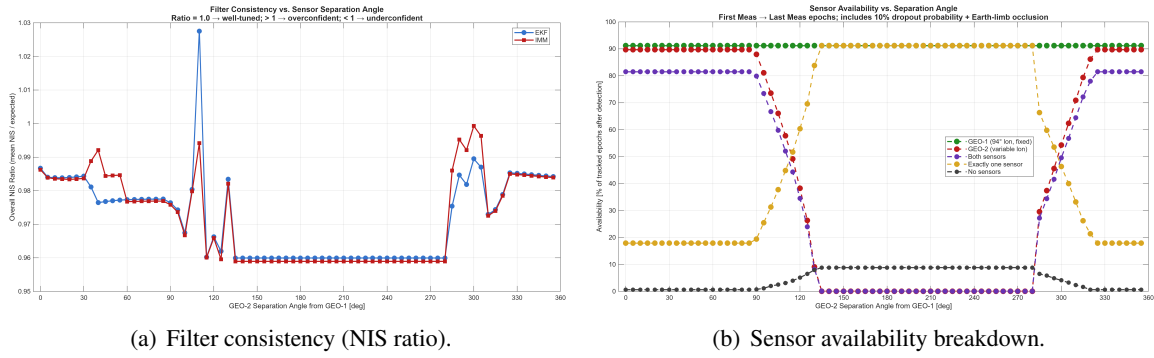


Figure 8. Sweep consistency and availability. Left: NIS ratio (mean NIS / expected NIS) stays within about 4.5% of unity, usually just below 0.98, with a narrow spike at the occlusion transition (110°) where measurement count fluctuates most rapidly. Right: availability breakdown showing when GEO-2 enters and exits the Earth-limb occlusion zone.

small (< 5 km) even in the single-sensor regime. The along-LOS (depth) component $e_{||}$ is what the single sensor cannot constrain.

Figure 9 (right) shows the mean along-LOS error for single-sensor epochs, together with the single-sensor epoch fraction, as functions of separation angle. In the dual-sensor regime, $e_{||}$ remains near zero because GEO-2 constrains depth directly. In the full-occlusion zone, $e_{||}$ saturates at 146 km (EKF) and 156 km (IMM), the equilibrium depth error achievable through dynamical parallax alone from a single GEO sensor on this trajectory.

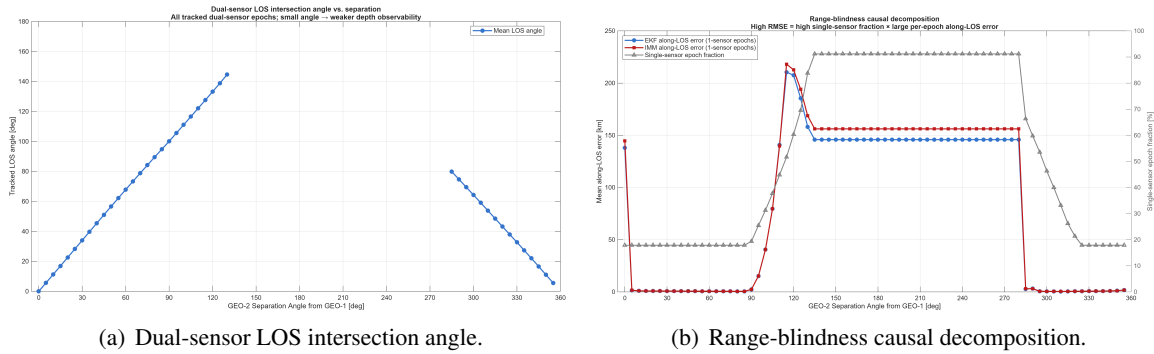


Figure 9. Observability geometry. Left: mean LOS intersection angle between the two sensors during dual-sensor epochs; the angle disappears in the occlusion zone (120° – 275°) where GEO-2 is Earth-blocked. Right: along-LOS (depth) error for single-sensor epochs overlaid with the single-sensor epoch fraction. The depth error peaks transiently to 210–218 km at occlusion onset before settling to the 146–156 km single-sensor equilibrium.

Occlusion-Onset Transient and Asymmetric Recovery

The range-blindness decomposition also explains the two asymmetric features of the RMSE curve in Figure 7.

Onset transient (90° – 120°): At these separations, GEO-2 is visible during the early boost arc but becomes occluded partway through the flight. During the dual-sensor phase, the filter converges to

a tight estimate with a small depth covariance $P_{\parallel} \ll \sigma_b^2$. When GEO-2 is lost, process noise begins inflating $P_{\parallel}$, but the true depth error grows faster than $P_{\parallel}$ widens—because the Kalman gain in the depth direction remains small, inherited from the still-tight dual-sensor covariance. The consequence is a transient: the true depth error runs ahead of the filter’s admitted uncertainty, overshooting to 210 km before the Kalman gain equilibrates and depth error settles to the 146 km single-sensor plateau. This overshoot is visible in both Figure 7 and Figure 9 (right).

Asymmetric recovery (275°–295°): When GEO-2 re-enters the FOV, recovery is nearly instantaneous rather than gradual. The reason is a fundamental asymmetry in Kalman filtering: after a long single-sensor phase, $P_{\parallel}$ has inflated to its single-sensor equilibrium—a large value in the depth direction. When GEO-2 returns, this large prior uncertainty produces a large Kalman gain precisely in the depth direction, and the filter corrects its depth estimate strongly from the first new triangulating measurement. Depth error collapses within a handful of epochs. There is no slow re-convergence because the filter’s high covariance already signals that it needs the depth measurement. It receives one, the gain is high, and the correction is large. Contrast this with the onset behavior, where the filter’s inherited confidence from the dual-sensor phase suppresses the Kalman gain and allows the error to grow uncorrected.

EKF vs. IMM plateau difference: The IMM plateau (156 km) exceeds the EKF plateau (146 km) by 10 km. In the single-sensor regime, the IMM mixes the boost and coast sub-filter states, whose acceleration priors disagree. This mixing injects additional noise into the depth estimate without a compensating measurement constraint, elevating the equilibrium depth error slightly.

CONCLUSION

This paper has demonstrated that, for the EKF, UKF, and IMM implementations studied here, sensor-placement geometry influences angles-only GEO IR tracking accuracy more strongly than the choice of filter architecture.

At the nominal 45° separation, the EKF and UKF each achieve 1.621 km 3D position RMSE and 0.096 km/s 3D velocity RMSE. The IMM performs marginally better, achieving 1.607 km 3D position RMSE and 0.094 km/s 3D velocity RMSE, with all three filters yielding a mean normalized innovation squared near the expected value. The EKF and UKF produce numerically near-identical results, confirming that the measurement model is not sufficiently nonlinear to warrant the additional complexity of sigma-point propagation. The IMM’s primary advantage over the EKF is a smooth, probabilistically-weighted boost-to-coast transition that avoids the one-step Singer-parameter discontinuity of the EKF’s hard acceleration-norm latch. Its tradeoffs are an 80–85 s detection lag before the coast model fully dominates and mild model-mixing velocity noise during boost.

For this particular trajectory, the sensor geometry sweep identifies an optimal separation band of 10°–85° within which dual-sensor triangulation reduces coast-phase RMSE by two orders of magnitude relative to single-sensor performance. Sensor separations that place GEO-2 behind the Earth’s limb drive RMSE to a 238–249 km plateau governed by the equilibrium depth uncertainty of a single angles-only observer. Filter consistency is maintained across all tested geometries (within about $\pm 4.5\%$ in NIS ratio), indicating that the Singer process-noise tuning is robust to large changes in measurement geometry.

The LOS error decomposition reveals that the along-LOS depth component is the sole source of the high single-sensor RMSE. The cross-LOS angular error remains small regardless of geometry. The characteristic transient overshoot at occlusion onset and the abrupt asymmetric recovery at

occlusion exit are both consequences of Kalman gain dynamics in the depth direction, where a previously confident filter loses its depth sensor and a covariance-inflated filter recovers it in a single update.

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NOTATION

$\mathbf{r}, \mathbf{v}, \mathbf{a}$ ECI position, velocity, augmented acceleration	$\mathbf{S}$ innovation covariance
$\mathbf{x}$ nine-dimensional filter state vector	τ, σ Singer correlation time and steady-state RMS
$\hat{\mathbf{x}}^+, \hat{\mathbf{x}}^-$ updated (posterior) and predicted (prior) state	μ_j IMM model probability for model j
$\mathbf{P}^+, \mathbf{P}^-$ updated and predicted error covariance	p_{ij} Markov transition probability, model i to j
$\mathbf{F}$ state transition matrix (EKF linearization)	Λ_j model likelihood
$\mathbf{Q}_c, \mathbf{Q}_d$ continuous- and discrete-time process noise covariance	ϵ_k normalized innovation squared (NIS) at epoch k
$\mathbf{H}$ measurement Jacobian	$\hat{\mathbf{l}}$ unit LOS vector from sensor to vehicle
$\mathbf{K}$ Kalman gain	$e_{\parallel}$ along-LOS (depth) error component
$\mathbf{R}$ measurement noise covariance	$\mathbf{\Pi}$ Markov transition matrix

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